

Paper 22.25 - MetaForge Toolkit Supplement

CQE-paper-22.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-22.25.md

Paper 22.25 - MetaForge Toolkit Supplement

Purpose

This supplement describes how to run and inspect the MetaForge materials pipeline. It supports Paper 22 but does not replace its proof.

Digital Route

Run:

```
`python production/formal-papers/CQE-paper-22/verify_metaforge_materials.py`
```

The expected status is ``pass_with_open_obligations``. The run writes

``metaforge_materials_receipt.json`` and checks material inventory, Pareto partner selection, ten-fold evaluation, seam proposal, production accounting, and additional material-pair replay.

Package Route

The promoted package lives at:

```
`production/lib-forge/CQECMPLX-MetaMaterial-Designer/meta_material_system`
```

The key source surfaces are:

- ``material_db.py`` for material records,
- ``pareto_partnering.py`` for partner scoring,
- ``fold_evaluation.py`` for ten-fold candidate transport,
- ``seam_detection.py`` for mitigation rows,
- ``production_energy.py`` for production estimates,
- ``meta_material_designer.py`` for the combined pipeline.

Analog Route

A reviewer can reproduce the candidate logic with cards. Each material is a card with lattice, property, gluon, and oloid fields. Pair cards by score, advance the chosen pair through ten fold cards, place an error-wall token when a fold creates a boundary issue, and then attach seam cards as mitigation rows. The analog version is accepted only if it preserves the same candidate status: not fabricated, not measured, and not promoted beyond its receipt.

Hidden-Guess Diagnostic

Training mode should hide the diagnostic label until after the reviewer chooses one:

- valid candidate,
- invalid material row,
- partner-score row,
- fold-accounting row,
- seam obligation,
- production estimate,
- invalid promotion.

The revealed answer teaches the boundary between a generated candidate and a tested material.